

MAT 320 Final Project - Marcus's Car

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Abstract

Introduction

This project looks at a long-term, real-life dataset that tracks the mileage and fuel fill-ups of a 2014 Subaru Forester. The records begin in 2014 when Marcus's grandparents bought the car and mainly used it for city driving, carefully writing down each fill-up in a small journal. In 2022, Marcus took over the vehicle and continued the journal, first using his vehicle for regular trips to Lakeland University and later, starting in 2025, for more frequent and longer drives related to his job (including travel to places like Brown Deer and Sturgeon Bay) and visits to his girlfriend in Manitowoc. Because the data were simply recorded as the car was used, this is an observational data set rather than an experiment, but it is consistent and detailed due to the careful record-keeping. Using this information, the goal is to examine how gas prices (dollars per gallon), fuel efficiency (miles per gallon), and overall driving patterns have changed over time, especially across these different phases of use, and to compare Marcus's driving habits with those of his grandparents.

The main research questions of this data focus on identifying trends and changes over time in:

- Cost of gasoline over time (dollars per gallon)
- Fuel efficiency over time (miles per gallon)
- Driving frequency and distance (miles driven across different time periods)

Method and Results

The analysis uses the data collected from the records kept in the journal from Marcus and his grandparents tracking the date, mileage, gallons, and cost of each car fill-up. Using this data, we have also calculated miles per gallon, dollars per gallon, ownership, and season of each data entry.

The libraries and data set we will be working with:

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(lubridate)

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union

library(ggplot2)

carData <- read.csv("MAT 320 Data.csv")
```

To get a basic understanding of the data we look at the following graphs and calculations.

```
entries <- nrow(carData)
totalDollars <- sum(carData$Cost, na.rm = TRUE)
totalMiles <- 156135
dollarsPerMile <- totalDollars / totalMiles
avgDollars <- sprintf("%.2f", mean(carData$DPG, na.rm = TRUE))
avgMileage <- sprintf("%.2f", mean(carData$MPG, na.rm = TRUE))

dollarsPerMile <- round(dollarsPerMile, 2)
totalDollars <- sprintf("%.2f", totalDollars)
```

Our data set spans from March 7th 2014 to April 25th 2026. Below are some interesting summary statistics:

- The total number of entries for our data set is 670.
- The total dollars spent on gas is \$15645.28.
- The total miles on the car is 156,135.
- This means that on average each mile cost approximately \$0.1.
- The total average dollars per gallon is \$2.63.
- The total average miles per gallon is 26.73.

First we will show the average Dollars per Gallon per year and average Dollars per Gallon per season.

Note that the seasons have been created as follows:

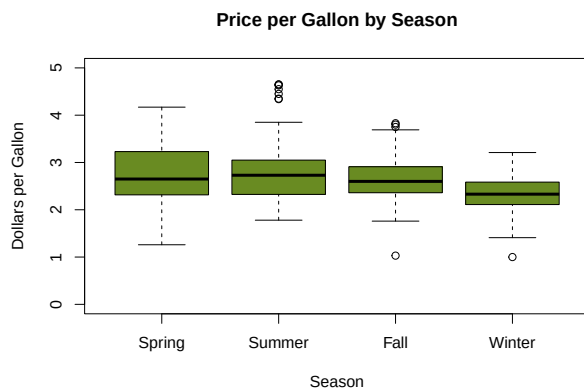
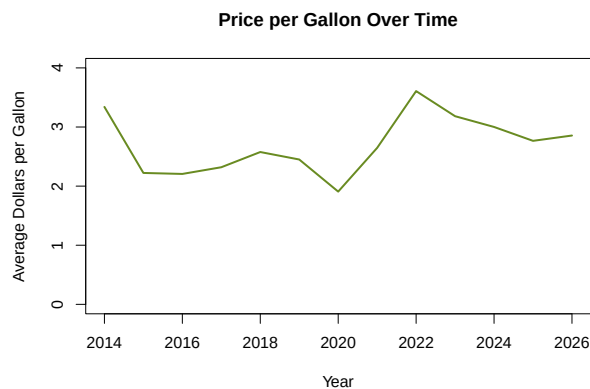
- 0 = Spring (March, April, May)
- 1 = Summer (June, July, August)

2 = Fall (September, October, November)
3 = Winter (December, January, February)

```
AvgDPGPerYear <- aggregate(DPG ~ Year, data = carData, FUN = mean)
```

```
plot(AvgDPGPerYear$Year, AvgDPGPerYear$DPG,  
     type = "l", col = "olivedrab4", lwd = 2,  
     main = "Price per Gallon Over Time",  
     xlab = "Year", ylab = "Average Dollars per Gallon",  
     ylim = c(0, 4))
```

```
boxplot(DPG ~ Season, data = carData,  
        col = "olivedrab4",  
        main = "Price per Gallon by Season",  
        xlab = "Season", ylab = "Dollars per Gallon",  
        ylim = c(0, 5),  
        names = c("Spring", "Summer", "Fall", "Winter"))
```



From these graphs we can see there may be a significant difference in gas prices between seasons. Lets find out by using a one-way ANOVA. First we need to check our assumptions.

```
spring <- carData$DPG[carData$Season == 0]  
summer <- carData$DPG[carData$Season == 1]  
fall   <- carData$DPG[carData$Season == 2]  
winter <- carData$DPG[carData$Season == 3]
```

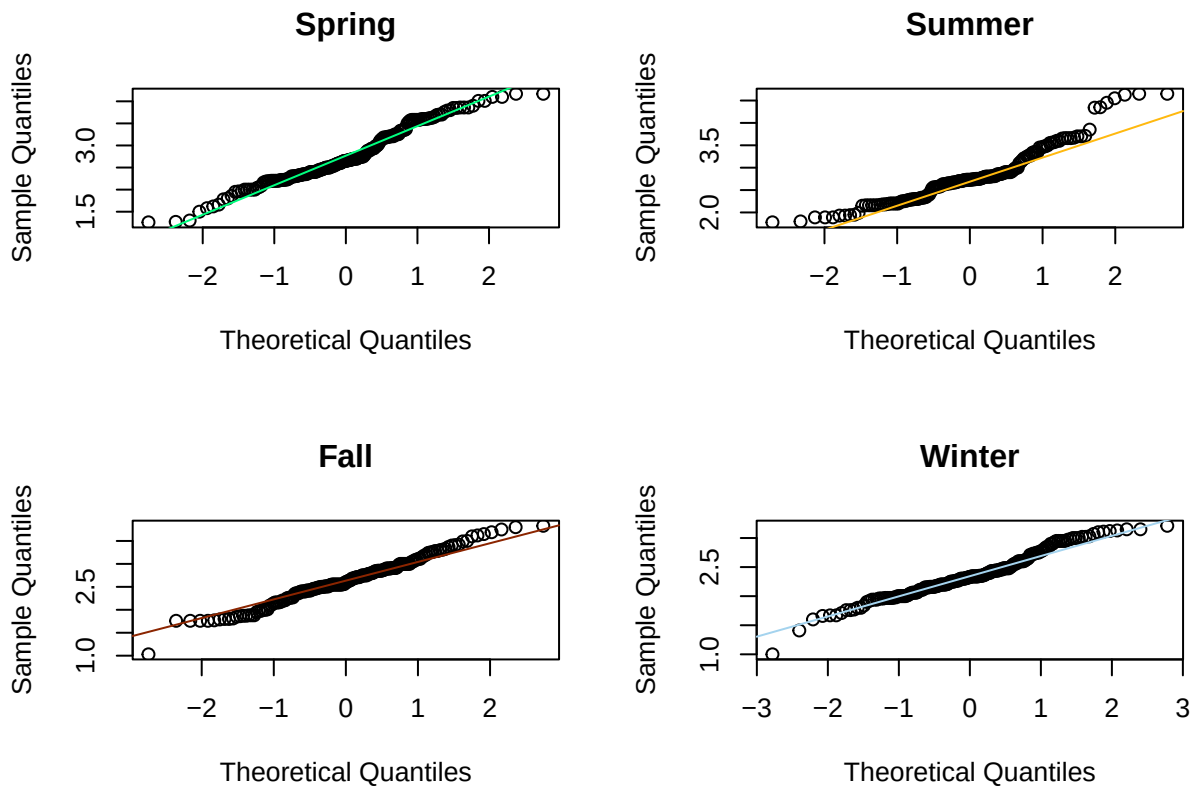
```
par(mfrow = c(2, 2))
```

```
qqnorm(spring, main = "Spring")  
qqline(spring, col = "springgreen")
```

```
qqnorm(summer, main = "Summer")  
qqline(summer, col = "darkgoldenrod1")
```

```
qqnorm(fall, main = "Fall")  
qqline(fall, col = "orangered4")
```

```
qqnorm(winter, main = "Winter")  
qqline(winter, col = "lightskyblue2")
```



```
sd_vals <- c(sd(spring), sd(summer), sd(fall), sd(winter))
max(sd_vals) / min(sd_vals)
```

```
## [1] NA
```

All assumptions are met since we have an independent simple random sample, the population is normally distributed as shown by the Q-Q plots, and we have similar standard deviations since the ratio of the largest to the smallest sample standard deviation is less than 2. It is appropriate to apply one-way ANOVA procedures to this data.

```
dollarsAnova <- aov(DPG ~ factor(Season), data = carData)
summary(dollarsAnova)
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## factor(Season)  3  21.75    7.25   24.98 2.52e-15 ***
## Residuals      665 193.00    0.29
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 1 observation deleted due to missingness
```

At the 5% significance level there is enough evidence to conclude that a difference exists among the mean price per gallon of gas during different seasons. The p-value is a lot less than 0.05 so we reject the null hypothesis of them being the same.

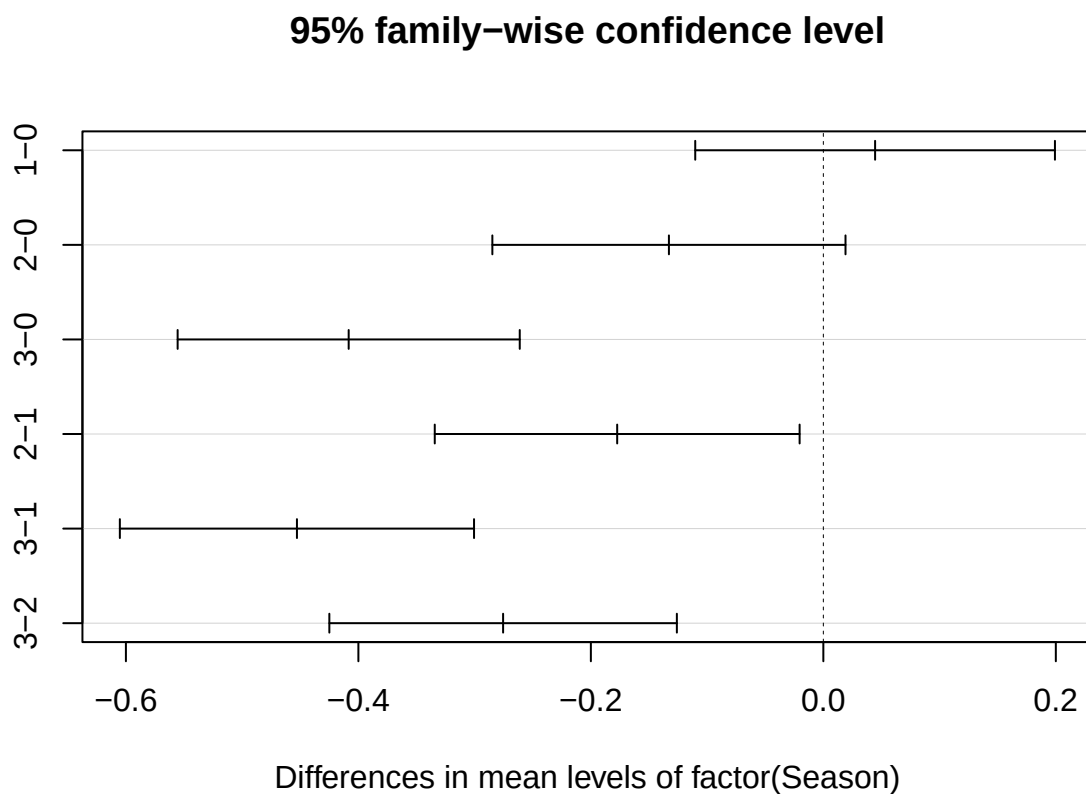
Next lets see which seasons are different.

```
TukeyHSD(dollarsAnova)
```

```
## Tukey multiple comparisons of means
```

```
##      95% family-wise confidence level
##
## Fit: aov(formula = DPG ~ factor(Season), data = carData)
##
## $`factor(Season)`
##           diff           lwr           upr      p adj
## 1-0  0.04451525 -0.1102169  0.19924742 0.8804933
## 2-0 -0.13284597 -0.2847550  0.01906305 0.1105064
## 3-0 -0.40835187 -0.5555099 -0.26119385 0.0000000
## 2-1 -0.17736121 -0.3343101 -0.02041232 0.0194755
## 3-1 -0.45286712 -0.6052222 -0.30051199 0.0000000
## 3-2 -0.27550590 -0.4249930 -0.12601881 0.0000150
```

```
plot(TukeyHSD(dollarsAnova))
```



There is enough evidence to show a difference in gas prices between fall and winter, summer and winter, summer and fall, and spring and winter. There is not enough evidence to show a difference in gas prices between spring and fall or spring and summer since these confidence intervals contain zero. However it is worth noting that the spring and fall interval just barely contains zero.

Next we will show the average Mileage per Gallon per year and average Mileage per Gallon per season.

```
AvgMPGPerYear <- aggregate(MPG ~ Year, data = carData, FUN = mean)

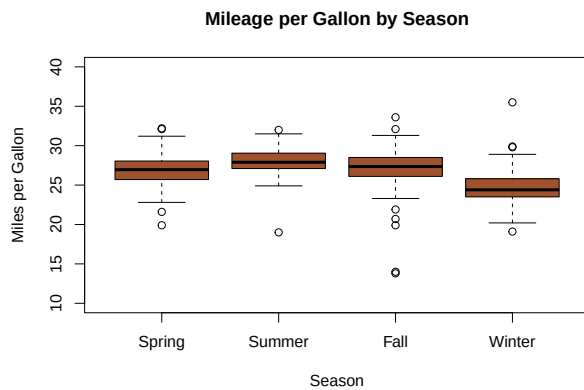
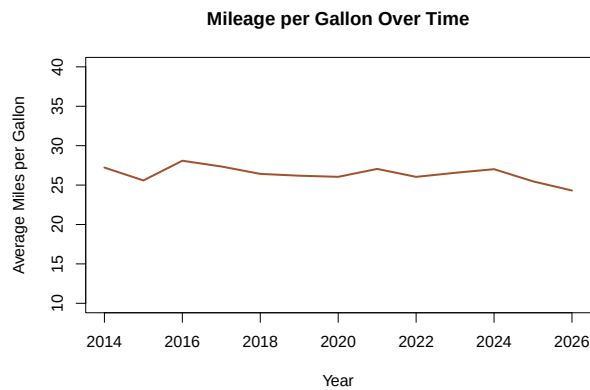
plot(AvgMPGPerYear$Year, AvgMPGPerYear$MPG,
     type = "l", col = "sienna", lwd = 2,
     main = "Mileage per Gallon Over Time",
     xlab = "Year", ylab = "Average Miles per Gallon",
```

```

ylim = c(10,40))

boxplot(MPG ~ Season, data = carData,
        col = "sienna",
        main = "Mileage per Gallon by Season",
        xlab = "Season", ylab = "Miles per Gallon",
        ylim = c(10, 40),
        names = c("Spring", "Summer", "Fall", "Winter"))

```



From these graphs we can see there may be a significant difference in gas mileage between seasons. Lets find out by using a one-way ANOVA. First we need to check our assumptions.

```

spring <- carData$MPG[carData$Season == 0]
summer <- carData$MPG[carData$Season == 1]
fall <- carData$MPG[carData$Season == 2]
winter <- carData$MPG[carData$Season == 3]

par(mfrow = c(2, 2))

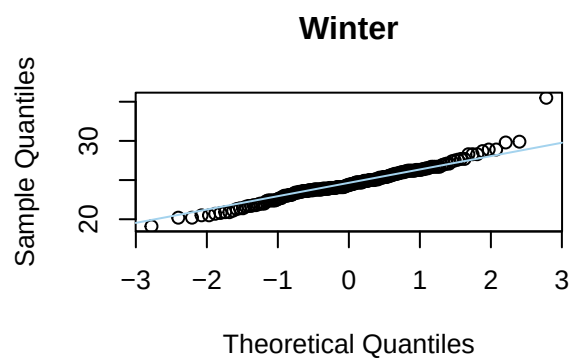
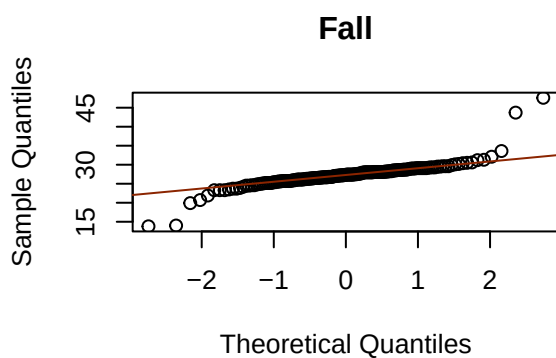
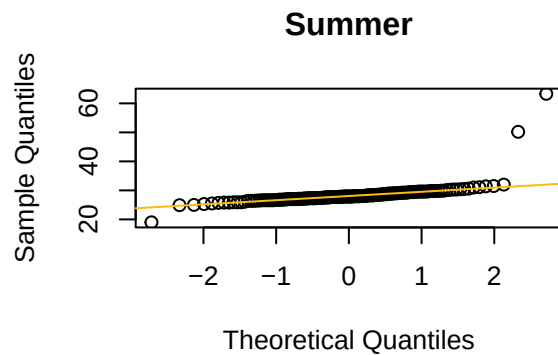
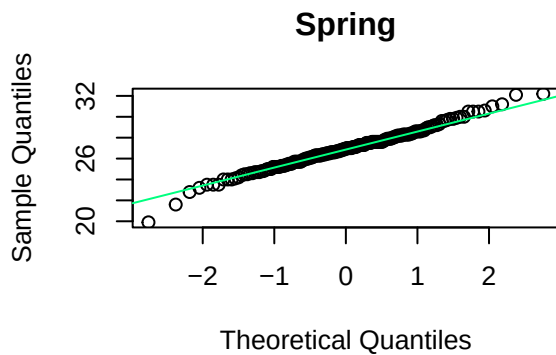
qqnorm(spring, main = "Spring")
qqline(spring, col = "springgreen")

qqnorm(summer, main = "Summer")
qqline(summer, col = "darkgoldenrod1")

qqnorm(fall, main = "Fall")
qqline(fall, col = "orangered4")

qqnorm(winter, main = "Winter")
qqline(winter, col = "lightskyblue2")

```



```
sd_vals <- c(sd(spring), sd(summer), sd(fall), sd(winter))
max(sd_vals) / min(sd_vals)
```

```
## [1] NA
```

All assumptions are met since we have an independent simple random sample, the population is normally distributed as shown by the Q-Q plots, and we have similar standard deviations since the ratio of the largest to the smallest sample standard deviation is less than 2. It is appropriate to apply one-way ANOVA procedures to this data. I would like to point out that the ratio is close to 2 being at 1.96158.

```
milesAnova <- aov(MPG ~ factor(Season), data = carData)
summary(milesAnova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## factor(Season)  3  1335   445.1    57.41 <2e-16 ***
## Residuals      665   5156     7.8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 1 observation deleted due to missingness
```

At the 5% significance level there is enough evidence to conclude that a difference exists among the mean mileage per gallon of gas during different seasons. The p-value is a lot less than 0.05 so we reject the null hypothesis of them being the same.

Next lets see which seasons are different.

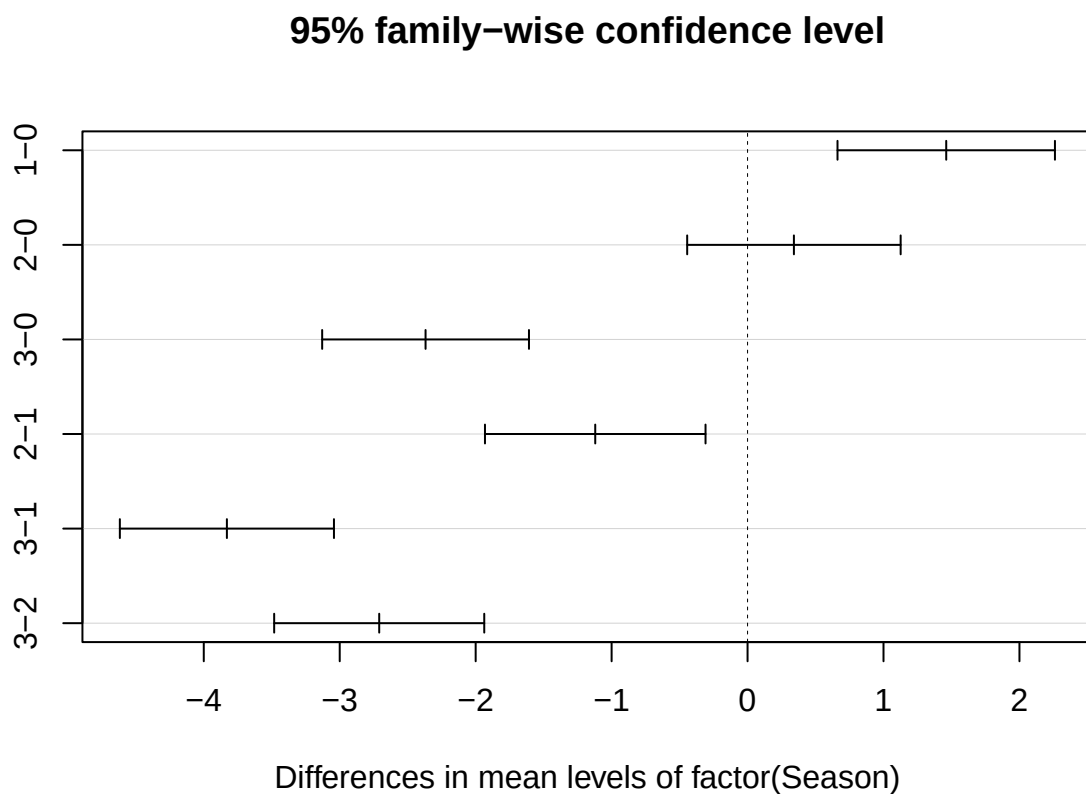
```
TukeyHSD(milesAnova)
```

```
## Tukey multiple comparisons of means
```



```
##      95% family-wise confidence level
##
## Fit: aov(formula = MPG ~ factor(Season), data = carData)
##
## $`factor(Season)`
##      diff      lwr      upr    p adj
## 1-0  1.4613913  0.6616139  2.2611688 0.0000182
## 2-0  0.3410128 -0.4441724  1.1261980 0.6781905
## 3-0 -2.3685216 -3.1291498 -1.6078933 0.0000000
## 2-1 -1.1203785 -1.9316138 -0.3091433 0.0022671
## 3-1 -3.8299129 -4.6174039 -3.0424219 0.0000000
## 3-2 -2.7095344 -3.4822011 -1.9368676 0.0000000
```

```
plot(TukeyHSD(milesAnova))
```



There is enough evidence to show a difference in gas mileage between fall and winter, summer and winter, summer and fall, and spring and winter. There is not enough evidence to show a difference in gas prices between spring and fall or spring and summer since these confidence intervals contain zero.

This is the same conclusion we came to about the differences in price per gallon.

Now we will look at the annualized miles based on the car's ownership status. The data has been annualized to show the number of miles driven per year under each status of ownership, and has been estimated based off of total dates since the ownership status of the car has changed mid-year. Note that the ownership status has been recorded as follows:

0 = Marcus's grandparents
 1 = Marcus before Amber
 2 = Marcus after Amber

The ownership status of the vehicle before and after Marcus met Amber also represents when he started regularly driving to Brown Deer and Sturgeon Bay for work.

```
carData$Date <- as.Date(carData$Date)

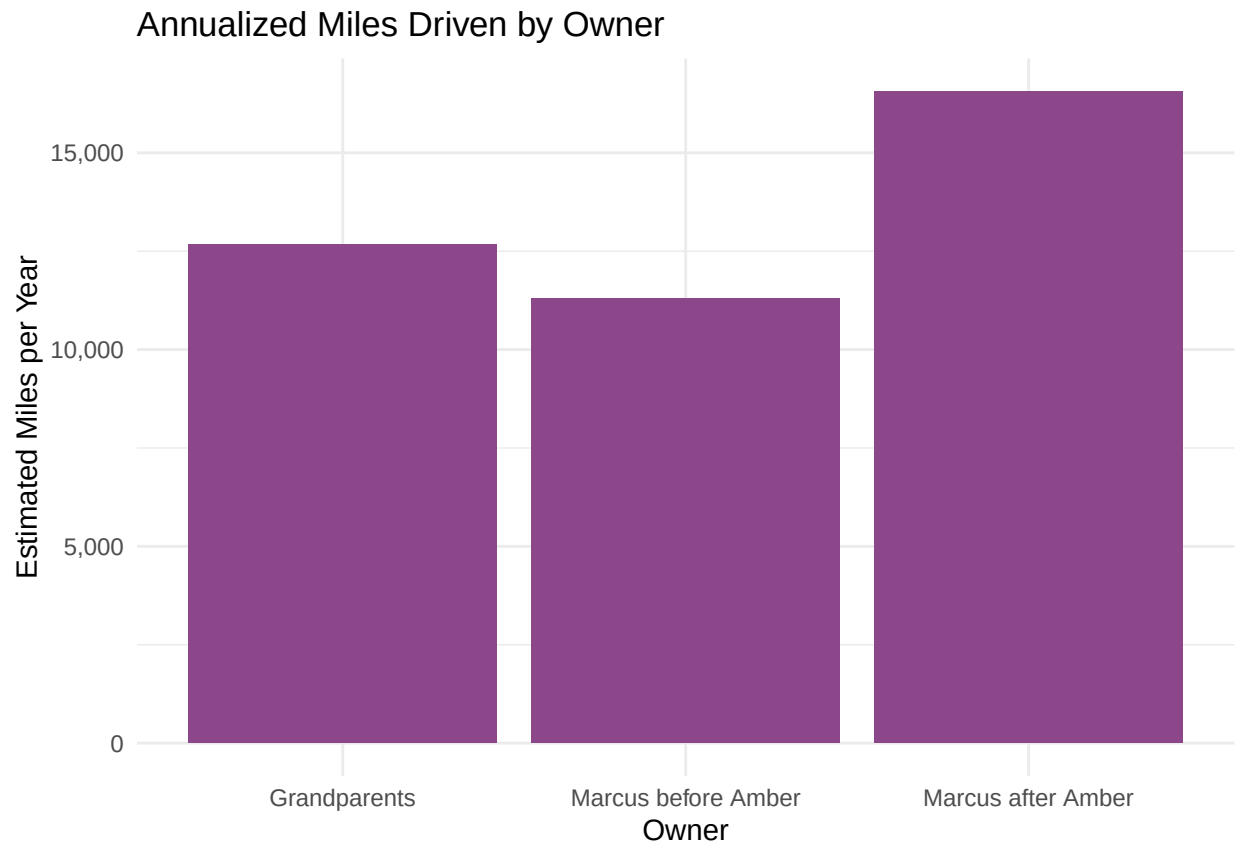
owner_miles <- carData %>%
  filter(!is.na(Milage)) %>%
  group_by(Owner) %>%
  summarise(
    first_date = min(Date),
    last_date = max(Date),
    total_days = as.numeric(last_date - first_date),
    first_milage = min(Milage),
    last_milage = max(Milage),
    total_miles = as.numeric(last_milage - first_milage),
    miles_per_day = total_miles / total_days,
    annualized_miles = miles_per_day * 365
  )
owner_miles

## # A tibble: 3 x 9
##   Owner first_date last_date total_days first_milage last_milage total_miles
##   <int> <date>      <date>      <dbl>      <int>      <int>      <dbl>
## 1     0 2014-03-07 2022-08-12    3080         574    107473    106899
## 2     1 2022-08-24 2024-12-15     844    107782    133926    26144
## 3     2 2024-12-27 2026-04-25     484    134168    156135    21967
## # i 2 more variables: miles_per_day <dbl>, annualized_miles <dbl>

carData_with_intervals <- carData %>%
  arrange(Owner, Date) %>%
  group_by(Owner) %>%
  mutate(
    days_diff = as.numeric(Date - lag(Date)),
    miles_diff = Milage - lag(Milage),
    miles_per_day = miles_diff / days_diff
  ) %>%
  filter(!is.na(miles_per_day), days_diff > 0) %>%
  ungroup()

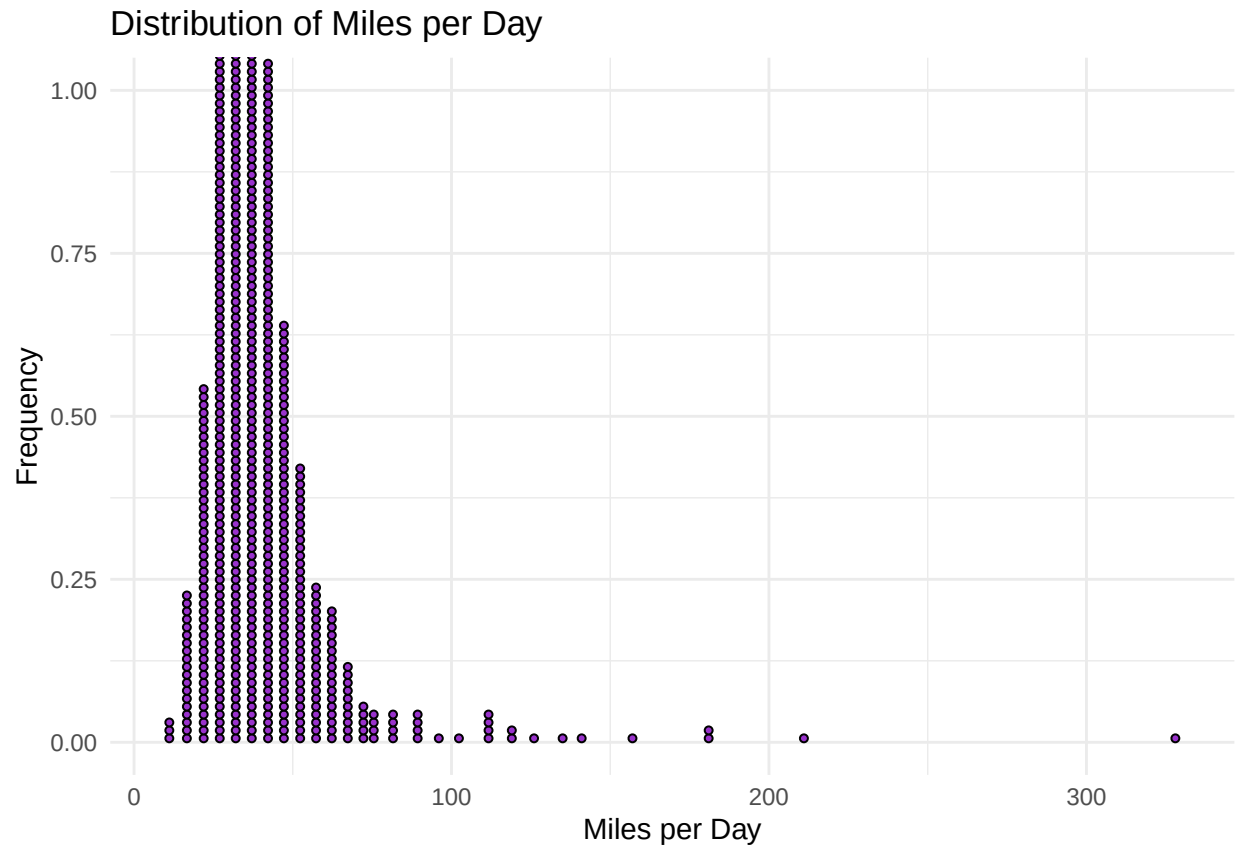
ggplot(owner_miles, aes(x = factor(Owner), y = annualized_miles)) +
  geom_col(fill = "orchid4") +
  labs(
    title = "Annualized Miles Driven by Owner",
    x = "Owner",
    y = "Estimated Miles per Year"
  ) +
  scale_x_discrete(
    labels = c(
      "0" = "Grandparents",
      "1" = "Marcus before Amber",
      "2" = "Marcus after Amber"
    )
  ) +
  scale_y_continuous(labels = scales::comma) +
```

```
theme_minimal()
```



From this bar graph we can see there may be a significant difference in driving distance between who was driving the car. Lets find out by using a Wilcoxon Signed-Rank Test. First we need to check our assumptions.

```
ggplot(carData_with_intervals, aes(x = miles_per_day)) +  
  geom_dotplot(  
    binaxis = "x",  
    fill = "darkorchid",  
    color = "black",  
    dotsize = 0.5,  
    binwidth = 5  
  ) +  
  labs(  
    title = "Distribution of Miles per Day",  
    x = "Miles per Day",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```



From looking at this dot plot we can see that our data is not symmetric so we can continue to do our test.

```
wilcox.test(
  miles_per_day ~ Owner,
  data = filter(carData_with_intervals, Owner %in% c(0, 1))
)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: miles_per_day by Owner
## W = 25235, p-value = 0.03426
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(
  miles_per_day ~ Owner,
  data = subset(carData_with_intervals, Owner %in% c(1, 2))
)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: miles_per_day by Owner
## W = 1322, p-value = 6.7e-10
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(
  miles_per_day ~ Owner,
  data = subset(carData_with_intervals, Owner %in% c(0, 2))
)
```

```
)
```

```
##  
## Wilcoxon rank sum test with continuity correction  
##  
## data: miles_per_day by Owner  
## W = 10214, p-value = 3.18e-09  
## alternative hypothesis: true location shift is not equal to 0
```

Conclusion